

## Article

# Analysis of Influencing Factors and Prediction of Provincial Energy Poverty in China Based on Explainable Deep Learning

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## Abstract

Energy poverty remains an important challenge for sustainable development in China, with pronounced regional disparities and evolving temporal dynamics that require accurate and interpretable prediction tools. This study develops a provincial panel-based framework that combines Energy Poverty Index (EPI) construction, SSA-LSTM prediction, SHAP-based model interpretation, and two-way fixed effects (TWFE) regression analysis. Using provincial data for China (2003–2022), we first construct a composite EPI with the entropy weight method, then apply a Sparrow Search Algorithm (SSA) to optimize LSTM hyperparameters for EPI forecasting. SHAP is used to interpret feature contributions to model-predicted EPI, and TWFE regression is used to provide complementary panel-data evidence on factor–EPI associations. The results show that the SSA-LSTM model outperforms benchmark machine learning and deep learning models in out-of-sample prediction performance. SHAP-based interpretation indicates that variables such as GDP, energy intensity, and power generation per capita contribute strongly to prediction variation, with notable regional heterogeneity. TWFE results are broadly consistent with several key patterns identified in the SHAP analysis. Overall, the proposed framework provides an accurate and interpretable provincial energy poverty prediction approach and offers a useful empirical reference for energy poverty monitoring and policy discussion.

**Keywords:** Sparrow Search Algorithm; SHAP; Long Short-Term Memory; two-way fixed effects; energy poverty index; influencing factors

## 1. Introduction

In the context of global energy transition and sustainable development [1], energy poverty has emerged as a critical issue that requires urgent attention worldwide [2]. This is especially true in China, where, despite sustained economic growth and the widening urban–rural gap, energy poverty remains a pressing concern in certain regions [3]. Energy poverty not only limits individuals’ access to basic living and production conditions but also undermines the economic development and environmental sustainability of society as a whole [4]. The Chinese government has made energy poverty alleviation a key policy objective, implementing a series of measures such as rural grid renovation [5] and energy accessibility initiatives [6], achieving significant progress. However, despite ongoing policy advancements, energy poverty continues to be particularly pronounced in certain regions, especially in the western and remote areas [7]. This highlights the need for continuous optimization of energy poverty alleviation strategies and more precise evaluation and



Academic Editor: Vladimír Bureš

Received: 28 January 2026

Revised: 8 March 2026

Accepted: 16 March 2026

Published: 17 March 2026

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forecasting. Therefore, conducting research on energy poverty, particularly through data-driven and accurate forecasting, not only provides scientific evidence for policy formulation but also supports more equitable energy distribution and the coordinated development of socio-economic systems [8]. Against this background, an important problem is how to construct a reliable provincial energy poverty index and develop an accurate, interpretable, and robust forecasting framework to identify the key factors associated with its spatiotemporal variation across regions in China.

Significant advancements have been made in energy poverty research, with scholars employing a variety of methods to assess, characterize, and predict its extent across different regions and populations [9,10]. Traditional evaluation methods primarily rely on techniques such as the entropy weight method [11] and the CRITIC method [12], which quantify the level of energy poverty by considering multiple dimensions, including energy consumption [13], energy supply [14], and energy investment [15]. Common evaluation indicators include per capita electricity consumption [14], natural gas supply rate [16], and the proportion of low-carbon energy in the energy mix [17]. These indicators provide a relatively clear reflection of the accessibility and cleanliness of energy services [18,19]. Such multidimensional index systems have laid an important foundation for measuring energy poverty and comparing regional disparities, especially in contexts where direct household-level deprivation data are limited [20,21]. However, these conventional methods also have limitations when addressing complex regional heterogeneity, dynamic temporal changes, and interactions among multi-dimensional socioeconomic and environmental factors [22]. In particular, static or semi-static composite evaluation approaches may be less capable of capturing nonlinear relationships and evolving patterns in energy poverty across regions and over time [23].

To overcome these challenges, an increasing number of studies in recent years have adopted data-driven forecasting methods, such as machine learning and deep learning models [24], to more accurately capture the trends and associated factors of energy poverty, particularly in identifying the relative importance of variables in prediction tasks [25]. Compared with traditional evaluation approaches, these methods have demonstrated considerable potential in improving prediction accuracy, enhancing model adaptability, and handling large-scale and high-dimensional datasets [26]. At the same time, the growing use of explainable machine learning techniques has further strengthened this research direction by enabling post hoc interpretation of model outputs and feature contributions [27,28]. This helps improve the transparency of prediction models and provides additional analytical support for understanding why certain variables are more influential in model predictions [29,30]. Nevertheless, existing studies still face challenges in balancing predictive performance and interpretability, ensuring temporal robustness in model evaluation, and clearly distinguishing model-based feature contribution from statistical association analysis [31,32]. These issues suggest the need for a more integrated framework that combines robust prediction, interpretable model explanation, and complementary empirical analysis when examining energy poverty in a dynamic regional context [33].

The prediction methods for energy poverty can generally be categorized into three main approaches. First, mathematical statistical models, such as multiple linear regression [34] and dynamic panel regression [35], are widely used in energy poverty analysis. These models predict energy poverty by establishing linear relationships between economic and social factors, with the advantage of simplicity, strong interpretability, and ease of understanding and application. However, these traditional statistical models often struggle to capture complex nonlinear relationships and high-dimensional interaction effects, and they impose relatively stringent assumptions on the data, which may result in insufficient prediction accuracy [36]. Secondly, machine learning methods, such as support vector

machines [37], random forests [38], and XGBoost [39], have gradually become mainstream in energy poverty prediction. These methods are effective in handling nonlinear relationships and high-dimensional data, and they do not rely on rigid assumptions, providing higher predictive accuracy [37–39]. However, machine learning models often lack sufficient interpretability, presenting a challenge for policymakers in terms of how to interpret the predictions generated by these models [39]. Lastly, hybrid optimization models, such as optimized deep learning models [40] and ensemble learning methods [41], combine multiple models to leverage the strengths of each, thereby enhancing prediction stability and accuracy. These methods are capable of addressing more complex scenarios and possess greater generalization ability, though they may lead to increased computational costs and pose difficulties in model tuning due to the complexity of the combined models [40,41].

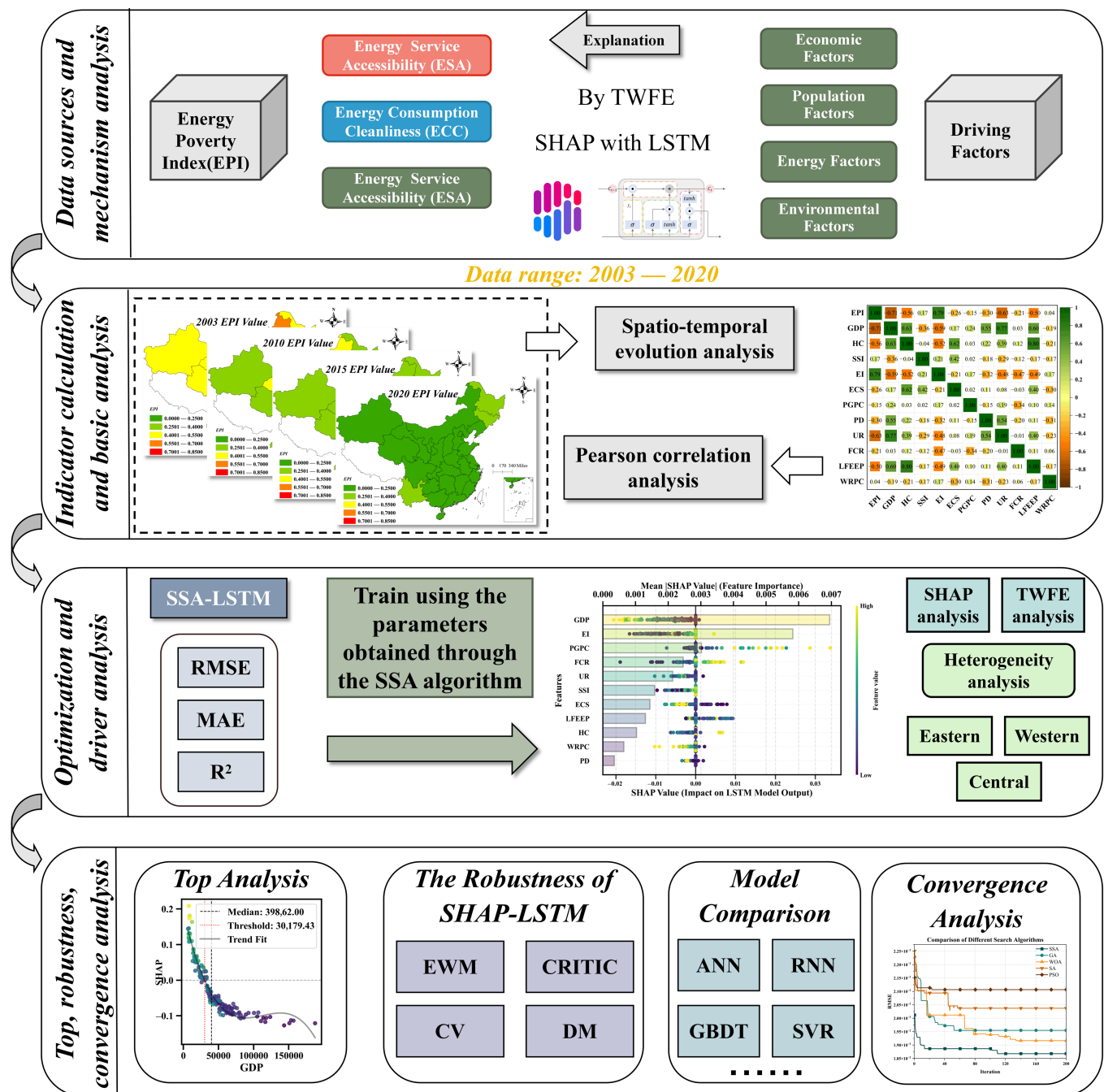
**Related work and incremental contribution:** Recent studies have increasingly applied explainable machine learning (e.g., SHAP) to energy poverty prediction, particularly using Chinese household-level microdata to interpret predictor relevance and heterogeneity (e.g., [42,43]). These works provide granular household-level insights on model interpretation and feature contribution patterns. A compact summary of representative energy poverty prediction studies and their main design choices is provided in Table 1. Relative to these household-focused studies, our incremental contribution lies in a province-year composite EPI forecasting setup with a strictly chronological split, the integration of SSA-optimized LSTM prediction with SHAP-based model interpretation, and complementary TWFE regression plus robustness and sensitivity checks to index construction and interpretation overlap.

**Table 1.** Analysis of relevant research.

| Ref. | Unit of Analysis | Models       | Interpretation | Key Focus                                          |
|------|------------------|--------------|----------------|----------------------------------------------------|
| [25] | Household        | ML           | Limited        | Household-level EP risk prediction                 |
| [42] | Household panel  | CNN          | SHAP           | Explainable DL for household EP                    |
| [43] | Household        | XGBoost      | SHAP           | Nonlinear predictor patterns in EP                 |
| [24] | Household        | Dual ML      | Discussed      | ML-based inference/prediction context              |
| [8]  | National/region  | ML ensembles | Limited        | Predictive modeling + determinants                 |
| [41] | National/region  | Multiple ML  | Limited        | Practical EP forecasting comparison                |
| [38] | Household/region | ML           | Decomposition  | Interpretable error decomposition                  |
| [34] | Household        | ANN          | Limited        | Predictive index for fuel poverty risk             |
| Ours | National/region  | SSA-LSTM     | SHAP + TWFE    | Robustness to EPI construction; sensitivity checks |

Building on the limitations of previous studies, this paper develops an integrated SSA-LSTM prediction and interpretation framework for provincial energy poverty analysis. First, the provincial Energy Poverty Index (EPI) is constructed using the entropy weight method. Second, a Long Short-Term Memory (LSTM) network is employed for time-series prediction, and the Sparrow Search Algorithm (SSA) is used to optimize key LSTM hyperparameters in order to improve model performance. Third, the optimized LSTM model is interpreted using SHAP to evaluate feature contributions to the model-predicted EPI and to improve the transparency of the prediction results. In addition, a two-way fixed effects (TWFE) panel regression is introduced as a complementary empirical analysis to examine the statistical associations between key explanatory variables and the EPI under province and year fixed effects.

Compared with conventional statistical models and single machine learning models, this framework is better suited to handling nonlinear relationships, multidimensional feature interactions, and regional heterogeneity in provincial panel data. By combining prediction, model interpretation, and panel-data regression analysis, the proposed framework provides a more systematic basis for understanding factor relevance in energy poverty analysis. The research framework is illustrated in Figure 1.



**Figure 1.** The research framework of this paper.

The contributions of this study can be summarized as follows:

- (1) Development of an SSA-LSTM-based provincial energy poverty prediction framework: This study develops an SSA-LSTM framework for provincial energy poverty prediction in China. By using SSA to optimize LSTM hyperparameters, the model improves predictive performance relative to benchmark models under the temporal evaluation setting adopted in this study.
- (2) Integration of SHAP-based model interpretation with panel-data regression analysis: In addition to SHAP-based interpretation of the SSA-LSTM prediction results, this study introduces a two-way fixed effects (TWFE) regression as a complementary analysis. This design helps distinguish model-based feature contribution patterns from panel-data statistical associations and improves the rigor of factor interpretation.



- (3) Provincial panel evidence and multi-model comparative evaluation: Using multi-year provincial panel data, this study constructs a composite EPI and conducts a comparative evaluation across multiple machine learning and deep learning models under a chronologically ordered training/validation/testing framework. The results provide provincial-level evidence on model performance and factor relevance, complementing existing household-level studies in the energy poverty literature.

## 2. Data Source and Methods

### 2.1. Data Sources and Variables Selection

The data used in this study were collected from the National Bureau of Statistics of China, the National Energy Administration, and annual provincial energy reports. The sample spans 2003–2022 and covers provincial indicators related to energy consumption, energy supply, energy investment, and socioeconomic and environmental conditions. Due to substantial data gaps and incomplete energy statistics, Tibet, Hong Kong, Macau, and Taiwan were excluded from the analysis. Before model construction, the raw data were systematically preprocessed. Specifically, we harmonized statistical calibers and units across sources, converted all variables into consistent numeric formats, and checked variable names and annual records for consistency during panel-data merging. We also screened for obvious data entry inconsistencies and corrected formatting issues (e.g., non-numeric symbols and inconsistent brackets or labels). After these cleaning and consistency checks, the final province–year panel used in the empirical analysis contains complete observations for the target variable and all explanatory variables. The resulting dataset combines temporal and cross-sectional information to support the measurement and prediction of energy poverty across Chinese provinces [13].

To evaluate energy poverty, this study constructs a multidimensional indicator system based on prior research [14]. The indicator system includes three dimensions: Energy Service Accessibility (ESA), Energy Consumption Cleanliness (ECC), and Household Energy Access (HEA) [15]. Importantly, the unit of analysis in this paper is the province-year, rather than individual households (or only rural households). Therefore, the index developed in this study is designed to measure a provincial composite energy poverty index (EPI), rather than a strictly rural energy poverty index.

At the provincial level, energy poverty reflects the overall conditions of energy service provision and household energy use across different settlement contexts. For this reason, the indicator system integrates both urban and rural dimensions. Specifically, ESA captures the accessibility and adequacy of basic energy consumption, supply, and investment; ECC reflects the cleanliness of energy use and energy structure; and HEA captures the availability of household energy-related facilities in both urban and rural settings. This integrated design is consistent with the objective of this study, which is to compare interprovincial and intertemporal differences in energy poverty conditions and to forecast provincial EPI dynamics using panel data.

Energy poverty refers to a condition in which individuals or groups cannot adequately access essential energy services required for basic living and production needs [17]. In this study, the EPI is constructed from the indicators listed in Table 2. The “Direction” column in Table 2 indicates the direction of contribution in index construction (i.e., whether a higher indicator value is associated with a lower or higher EPI score after standardization). For example, per capita electricity consumption and per capita natural gas consumption are treated as negative-direction indicators because, in general, higher access to these services is associated with lower energy poverty [16]. By contrast, per capita SO<sub>2</sub> emissions from residential activities are treated as a positive-direction indicator because higher emissions indicate less clean household energy use and, correspondingly, more severe energy poverty.

**Table 2.** Indicators for measuring provincial composite energy poverty index.

| Category                             | Indicator                   | Measurement                                                                                               | Average Value | Standard Deviation | Direction |
|--------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------|---------------|--------------------|-----------|
| Energy Service Accessibility (ESA)   | Energy consumption          | Per capita electricity consumption (kWh/person)                                                           | 4102.546      | 2934.403           | Negative  |
|                                      |                             | Per capita natural gas consumption (m <sup>3</sup> /person)                                               | 164.3585      | 187.6164           | Negative  |
|                                      | Energy supply               | Urban natural gas penetration rate (%)                                                                    | 88.69091      | 12.73225           | Negative  |
|                                      |                             | Urban per capita natural gas supply (m <sup>3</sup> )                                                     | 418.5563      | 1124.229           | Negative  |
|                                      | Energy investment           | Fixed asset investment in state-owned electricity/steam/hot water production and supply (100 million RMB) | 226.3922      | 172.4800           | Negative  |
| Energy Consumption Cleanliness (ECC) | Low-carbon energy structure | Share of non-thermal power generation (%)                                                                 | 0.264688      | 0.257143           | Negative  |
|                                      | Pollution from energy use   | Per capita SO <sub>2</sub> emissions from residential activities (tons)                                   | 0.001984      | 0.002507           | Positive  |
| Household Energy Appliances (HEA)    | Urban energy facilities     | Refrigerators per 100 urban households (units/100 households)                                             | 95.84461      | 7.855619           | Negative  |
|                                      |                             | Air conditioners per 100 urban households (units/100 households)                                          | 94.32699      | 72.53934           | Negative  |
|                                      | Rural energy facilities     | Range hoods per 100 rural households (units/100 households)                                               | 18.02621      | 19.86083           | Negative  |
|                                      |                             | Per capita coverage area of rural solar water heaters (m <sup>2</sup> )                                   | 0.134675      | 0.186392           | Negative  |

Because the EPI is constructed at the provincial composite level and includes both urban- and rural-related components, subsequent analyses involving variables such as the urbanization rate (UR) should be interpreted as predictive associations with model-predicted EPI outcomes, rather than direct causal mechanisms. To reduce the risk that urban-related EPI components mechanically amplify the apparent importance of UR, the revised manuscript further reports sensitivity analyses using alternative EPI specifications in the robustness section (e.g., excluding or down-weighting urban-specific components).

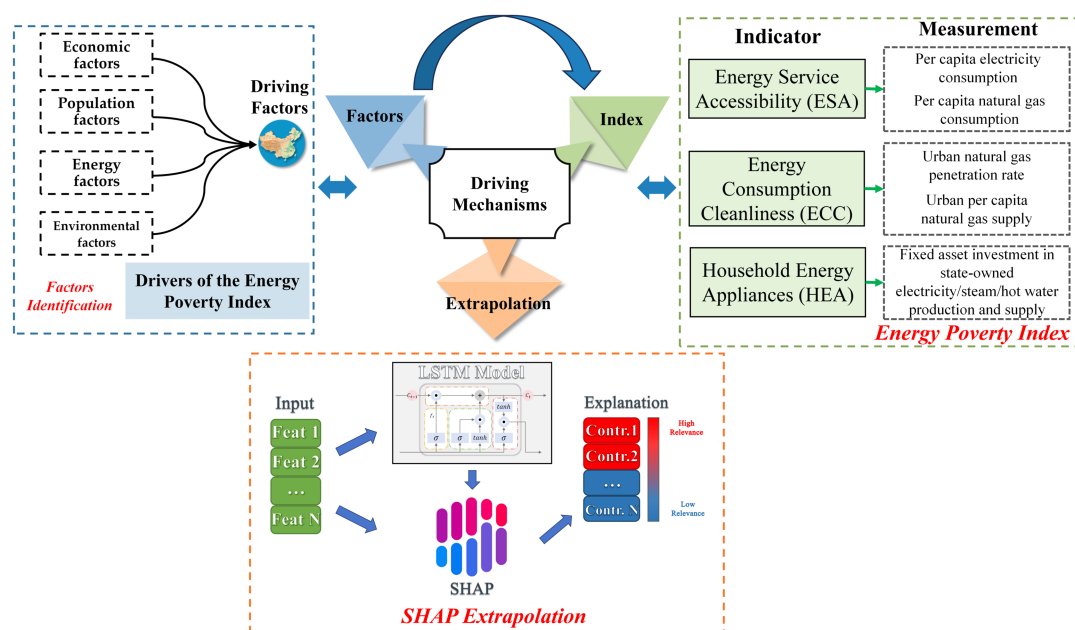
The descriptive statistics in Table 2 (pooled province-year observations) indicate substantial interprovincial heterogeneity in energy consumption, supply, and investment conditions. For example, the average per capita electricity consumption is 4102.55 kWh/person, with a standard deviation of 2934.40, indicating large differences in energy service accessibility across provinces. Similar disparities can be observed in natural gas consumption and supply indicators, which further reflect uneven energy infrastructure and service provision across regions. Building on the EPI construction, Table 3 presents the exogenous variables used in the prediction model and provides the background covariates for subsequent forecasting and SHAP-based interpretation [20,22].

Table 3 summarizes the exogenous variables used for energy poverty prediction, including economic factors (e.g., GDP per capita, household consumption, and the share of secondary industry), population factors (e.g., population density and urbanization rate), energy factors (e.g., energy intensity, energy consumption share, and power generation per capita), and environmental factors (e.g., forest coverage rate, local fiscal expenditure on environmental protection, and water resources per capita). The mean values and standard deviations of these variables indicate substantial interprovincial heterogeneity in economic conditions, demographic structure, energy-related characteristics, and environmental re-

source endowments. For example, the average GDP per capita is 43,442.5 CNY/person, with a standard deviation of 31,514.3, suggesting considerable differences in economic development levels across provinces. Similar dispersion is also observed for household consumption, population density, and water resources per capita, reflecting strong regional variation in socioeconomic and natural-resource conditions. These exogenous variables provide the key input features for the prediction model and the basis for subsequent SHAP-based interpretation. Based on these variables, Figure 2 presents the conceptual pathways through which these factors are associated with energy poverty [44].

**Table 3.** Exogenous variables used for energy poverty prediction.

| Factors               | Variable                                             | Symbols | Units                   | Average Value | Standard Deviation |
|-----------------------|------------------------------------------------------|---------|-------------------------|---------------|--------------------|
| Economic factors      | GDP per capita                                       | GDP     | CNY/person              | 43,442.5      | 31,514.3           |
|                       | Household consumption                                | HC      | 100 million CNY         | 7740.8        | 8373.3             |
|                       | Share of secondary industry                          | SSI     | %                       | 0.4231        | 0.0830             |
| Population factors    | Population density                                   | PD      | persons/km <sup>2</sup> | 448.03        | 660.24             |
|                       | Urbanization rate                                    | UR      | %                       | 53.940        | 14.534             |
| Energy factors        | Energy intensity                                     | EI      | ton/10 <sup>4</sup> CNY | 1.0801        | 0.7177             |
|                       | Energy consumption share                             | ECS     | %                       | 0.0337        | 0.0235             |
|                       | Power generation per capita                          | PGPC    | kWh/person              | 0.4503        | 0.4176             |
| Environmental factors | Forest coverage rate                                 | FCR     | %                       | 33.2726       | 17.941             |
|                       | Local fiscal expenditure on environmental protection | LFEEP   | 100 million CNY         | 108.151       | 96.539             |
|                       | Water resources per capita                           | WRPC    | m <sup>3</sup> /person  | 2135.1        | 2531.9             |



**Figure 2.** Diagram of the influencing factors of energy poverty.

## 2.2. Calculating Energy Poverty Index Using the Entropy Weight Method

In this study, we employ the Entropy Weight Method (EWM) to calculate the Energy Poverty Index (EPI) at the provincial level in China. The Entropy Weight Method is an objective weighting technique based on information entropy, which assigns different weights to indicators based on their variability (information content). This method is highly objective, as it does not rely on expert judgment and can effectively avoid the biases associated with subjective weighting [45].

Initially, the raw data must undergo standardization to eliminate the impact of differing units of measurement [45]. The standardized data can be calculated using the following formula

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}, \quad (1)$$

where  $x_{ij}$  represents the raw data of the  $j$ -th indicator for the  $i$ -th province, and  $x'_{ij}$  is the standardized data.  $\min(x_j)$  and  $\max(x_j)$  represent the minimum and maximum values of the  $j$ -th indicator across all provinces, respectively.

Next, we calculate the weight of each indicator using the following formula [45]

$$p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^m x'_{ij}}, \quad (2)$$

where  $p_{ij}$  represents the weight of the  $j$ -th indicator for the  $i$ -th province, and  $m$  is the total number of samples, i.e., the number of provinces 30.

Based on these weights, we compute the information entropy for each indicator, which measures the variability of the indicator [45]. The entropy is calculated as follows

$$e_j = -\frac{1}{\ln(m)} \sum_{i=1}^m p_{ij} \ln(p_{ij}), \quad (3)$$

where  $e_j$  represents the entropy value for the  $j$ -th indicator. A higher entropy value indicates less information, meaning the indicator contributes less to the total energy poverty index.

Finally, the weight for each indicator is computed from the information entropy using the following formula [45]

$$w_j = \frac{1 - e_j}{\sum_{j=1}^n (1 - e_j)}, \quad (4)$$

where  $w_j$  represents the weight of the  $j$ -th indicator, and  $n$  is the total number of indicators. Smaller entropy values correspond to larger weights, indicating that the indicator contributes more to the overall energy poverty index.

Once the weights for each indicator are obtained, the Energy Poverty Index (EPI) can be calculated by a weighted sum of the standardized indicators:

$$EPI = \sum_{j=1}^n w_j x'_{ij}, \quad (5)$$

where  $x'_{ij}$  is the standardized value of the  $j$ -th indicator and  $w_j$  is its corresponding weight. The resulting EPI reflects the overall level of energy poverty across the provinces.

### 2.3. SSA-LSTM Model Construction

#### 2.3.1. Sparrow Search Algorithm

This study utilizes the Sparrow Search Algorithm (SSA) to optimize the model for predicting energy poverty at the provincial level in China. SSA is a swarm intelligence optimization algorithm that simulates the foraging behavior of sparrows, inspired by the natural phenomenon in which sparrow groups cooperate and compete to find the optimal solution during food search. The advantage of SSA lies in its ability to thoroughly explore the solution space and improve search efficiency by combining local and global information. Compared to traditional optimization algorithms, SSA exhibits strong global search capability, especially in solving complex high-dimensional optimization problems [46].

The core idea of SSA is to solve optimization problems by simulating the foraging behavior of sparrow groups. In SSA, sparrow individuals are divided into two categories: leader sparrows and follower sparrows. Leader sparrows are responsible for guiding the

entire group through the search space, while follower sparrows rely on the position of the leader sparrow to conduct local searches. SSA iteratively updates the positions of both leader and follower sparrows to find the optimal solution to the optimization problem [47].

In SSA, the leader sparrow guides the group forward, and its position update rule is as follows [46]

$$X_{leader} = X_{leader} + \alpha(X_{best} - X_{leader}), \quad (6)$$

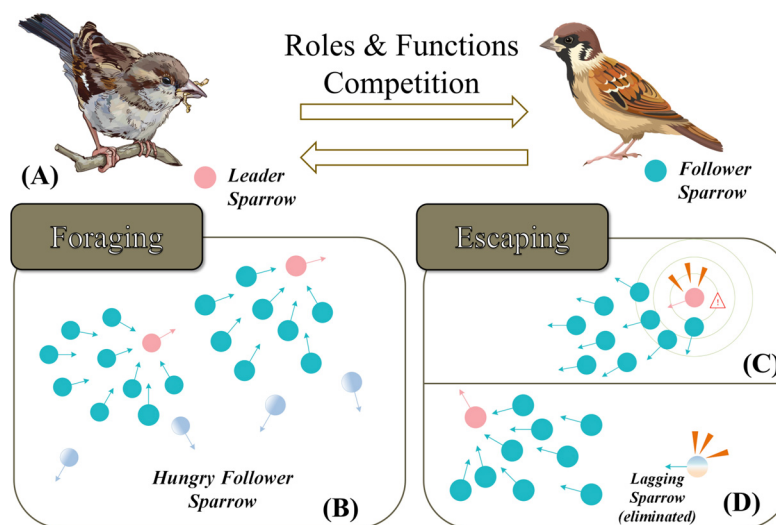
where  $X_{leader}$  represents the position of the leader sparrow,  $\alpha$  is a constant controlling the step size, and  $X_{best}$  is the current best solution in the group.

For follower sparrows, their position updates depend on the state of the leader sparrow as well as other sparrows in the group. The update rule for follower sparrows is as follows [47]

$$X_{follower} = X_{follower} + \beta(X_{leader} - X_{follower}) + \gamma(X_{random} - X_{follower}), \quad (7)$$

where  $X_{follower}$  represents the position of the follower sparrow,  $\beta$  and  $\gamma$  are constants controlling the influence of the leader sparrow and the random jump, respectively, and  $X_{random}$  represents the position of a randomly selected sparrow.

SSA continuously adjusts the positions of the leader and follower sparrows to conduct both global and local searches. In each iteration, SSA evaluates the fitness of the current position, which corresponds to the objective function value, and adjusts the search strategy based on changes in fitness. The algorithm terminates when the stopping criterion is met, which in this case is the maximum number of iterations, and the optimal solution is output. The algorithm framework is illustrated in Figure 3.



**Figure 3.** Sparrow search algorithm framework. (A) Two kinds of units in sparrow society: producers and scroungers, (B) starving scroungers for foraging, (C) aversion behavior and (D) straggler for aversion.

In terms of computational complexity, let  $P$  denote the population size,  $T$  the maximum number of iterations,  $D$  the dimension of the search space (i.e., the number of hyperparameters to be optimized), and  $C_f$  the computational cost of one fitness evaluation. In each iteration, SSA mainly involves position updates for leader and follower sparrows, population-level fitness evaluation, and ranking/selection operations. Therefore, the per-



iteration complexity can be expressed as  $O\left(T\left(PD + P\log P + PC_f\right)\right)$ , and the overall time complexity of SSA is approximately [47]:

$$O\left(T\left(PD + P\log P + PC_f\right)\right). \quad (8)$$

In the present study, the fitness function is defined as the validation RMSE of the LSTM model, so the dominant computational cost comes from repeated model training/evaluation (i.e.,  $C_f$ ). As a result, the practical computational burden of SSA-LSTM is primarily determined by the population size, the number of iterations, and the cost of LSTM training under each candidate parameter set. To balance optimization quality and computational efficiency, we further compare different search algorithms and conduct a population-size sensitivity analysis in Section 4.4.

### 2.3.2. Long Short-Term Memory (LSTM) Neural Network Model

This study utilizes Long Short-Term Memory (LSTM) networks to model and predict the time series of energy poverty index (EP) at the provincial level in China. Compared to traditional recurrent neural networks, LSTM effectively mitigates the vanishing gradient problem through its gating structure, enabling it to capture the dynamic evolution of energy poverty across different time scales. The LSTM layer updates the current state  $h_t$  and cell state  $c_t$  through three gate mechanisms: the forget gate, input gate, and output gate.

The forget gate is responsible for filtering historical memories. Its core function is to adaptively decide which information from the cell state  $C$  should be retained, weakened, or discarded based on the current input  $x_t$  and the previous hidden state  $h_{t-1}$ . This helps avoid the accumulation of redundant or outdated information during time propagation, improving the model's robustness to long-term dependencies and structural changes [48]. Its calculation is given by:

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right), \quad (9)$$

where  $f_t$  represents the activation vector of the forget gate, determining which information from the previous cell state needs to be forgotten,  $h_t$  is the current hidden state, and  $W$  and  $b$  are weight matrices and bias terms used in the calculations for the forget gate, input gate, candidate cell state, and output gate, respectively. The sigmoid function  $\sigma$  compresses values between 0 and 1, indicating whether the model should forget this information (closer to 0 means more likely to forget).

The input gate controls the extent to which new information is written into the cell state, deciding which parts of the current input features should be encoded and injected into the cell state. Specifically, the input gate uses the activation vector  $i_t$  to weight the candidate cell state  $\tilde{C}_t$ , performing selective memory of the new information. This allows the LSTM to suppress the writing of ineffective information when facing noise disturbances or short-term fluctuations while quickly learning key features when structural changes occur [49]. Its calculation is given by

$$\begin{cases} i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \end{cases} \quad (10)$$

where  $i_t$  is the activation vector of the input gate, deciding which new information will be added to the cell state, and  $\tilde{C}_t$  is the candidate cell state containing potential new information to be added to the current cell state. The hyperbolic tangent function  $\tanh$  modulates the nonlinearity of the information, with outputs between  $-1$  and  $1$ .

In the model, the cell state is updated by merging the historical memory retained by the forget gate with the new information introduced by the input gate. This results in

an accumulated representation of the long-term dependency characteristics of the energy poverty series, providing a stable memory carrier for subsequent state propagation and prediction [48]. Its calculation is

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \quad (11)$$

where  $C_t$  is the current cell state, representing an accumulated representation of past information, updated by considering both forgotten and new information, and  $\odot$  denotes element-wise multiplication.

Finally, the output gate extracts the most contributing information from the updated cell state  $C_t$  to generate the hidden state  $h_t$ . The output gate uses the activation vector  $o_t$  to determine which information from the cell state participates in subsequent calculations, enabling effective expression of the current prediction task while retaining long-term memory [50]. Its calculation is

$$\begin{cases} o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t = o_t \odot \tanh(C_t) \end{cases}, \quad (12)$$

where  $o_t$  is the activation vector of the output gate, determining which information from the cell state will be used to compute the current hidden state  $h_t$ . The SSA-LSTM model framework is shown in Figure 4.

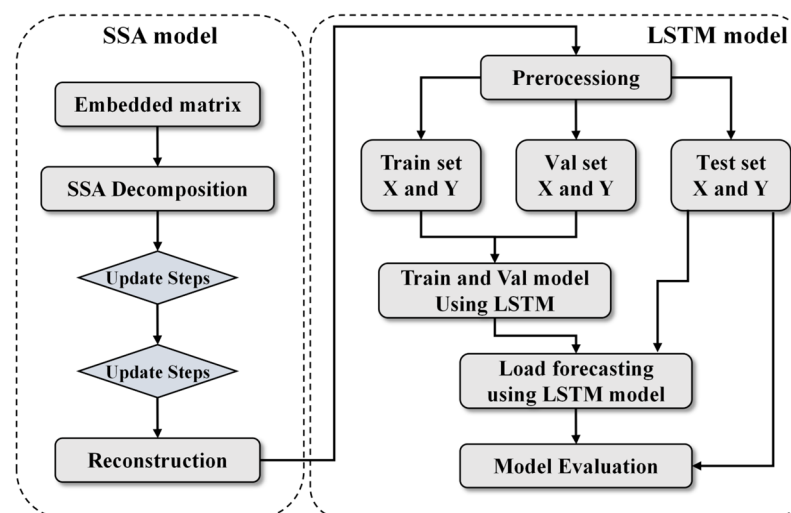


Figure 4. SSA-LSTM model framework.

## 2.4. Influencing Factor Analysis and Model Interpretation

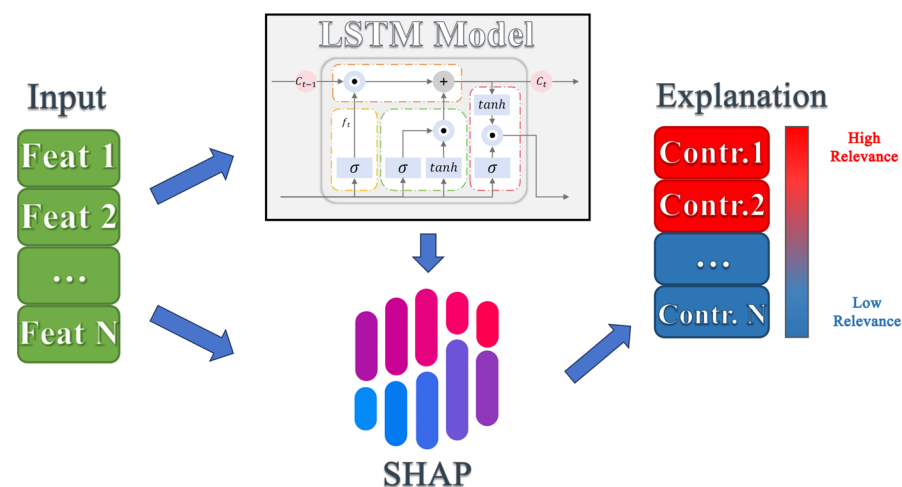
### 2.4.1. Influencing Factor Analysis and Model Interpretation

In this study, SHAP (SHapley Additive exPlanations) values are used to interpret the SSA-LSTM prediction model and identify the relative contribution of input features to model-predicted energy poverty levels at the provincial level in China. SHAP is derived from the Shapley value in cooperative game theory, and its core idea is to allocate the contribution of each feature to the model prediction by averaging its marginal contribution across all possible feature combinations [50]. This provides a unified framework for interpreting feature importance and direction in nonlinear prediction models. The calculation formula for SHAP values is as follows:

$$SHAP(f, x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(N - |S| - 1)!}{N!} [f(S \cup \{j\}) - f(S)], \quad (13)$$

where  $f(S)$  represents the prediction value of the feature set  $S$ ,  $S \cup \{j\}$  is the feature set  $S$  augmented by feature  $j$ ,  $|S|$  is the size of the set  $S$ , and  $N$  is the total number of features. SHAP values calculate each feature's contribution to the final prediction by considering the contributions of all feature combinations. Based on the size and sign of the SHAP values, we can identify which features have a significant positive or negative influence on the prediction of energy poverty.

Through SHAP value analysis, we gain an in-depth understanding of the contribution of various features to the energy poverty index, such as per capita regional GDP, energy consumption intensity, and forest coverage, among others. This method not only quantifies the contribution of each feature but also effectively explains the model's prediction results, enhancing the transparency and interpretability of the model. The framework for this method is shown in Figure 5.



**Figure 5.** SHAP-LSTM model framework.

#### 2.4.2. Two-Way Fixed Effects Regression for Associated-Factor Analysis

To complement the SHAP-based model interpretation and provide panel-data statistical evidence on factor–EPI relationships, this study further employs a two-way fixed effects (TWFE) regression model. The TWFE model controls for unobserved time-invariant provincial heterogeneity and common year shocks, thereby allowing a more rigorous examination of the associations between key explanatory variables and the provincial energy poverty index.

The baseline specification is as follows [37]:

$$EPI_{it} = \alpha + \beta' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (14)$$

where  $EPI_{it}$  denotes the provincial energy poverty index for province  $i$  in year  $t$ ,  $X_{it}$  is a vector of explanatory variables (e.g., economic, demographic, energy, and environmental factors),  $\mu_i$  represents province fixed effects,  $\lambda_t$  represents year fixed effects, and  $\varepsilon_{it}$  is the error term. In the TWFE analysis, the baseline specification uses lagged explanatory variables  $X_{i,t-3}$  to align the timing structure with the prediction framework; we additionally report results for  $X_{i,t-1}$  and  $X_{i,t-2}$  as robustness checks.

The two-way fixed effects regression is used to assess whether the directional associations suggested by the SHAP-based model interpretation are statistically consistent in the panel-data setting. However, we note that the TWFE results should be interpreted as associational evidence rather than definitive causal effects, unless additional identification strategies are introduced. In this study, the TWFE analysis is intended to provide

complementary statistical support for interpreting the factors related to provincial energy poverty variation.

### 3. Results

#### 3.1. EPI Calculation Results

We first calculate the Energy Poverty Index (EPI) for each province using the EPI indicators presented in Table 2. The years 2003, 2010, 2015, and 2020 were selected as representative years for display. These years were chosen because they are significant milestones: 2003 serves as the baseline year, reflecting the initial state of energy poverty; 2010 and 2015 correspond to key moments in economic transformation and energy structure adjustment; and 2020 marks an important year in China's battle against poverty, showcasing the most recent changes in energy poverty. The heat maps generated from the data for these years, shown in Figure 6, illustrate the distribution of EPI values and provide a visual representation of the spatiotemporal evolution of energy poverty across provinces [51].

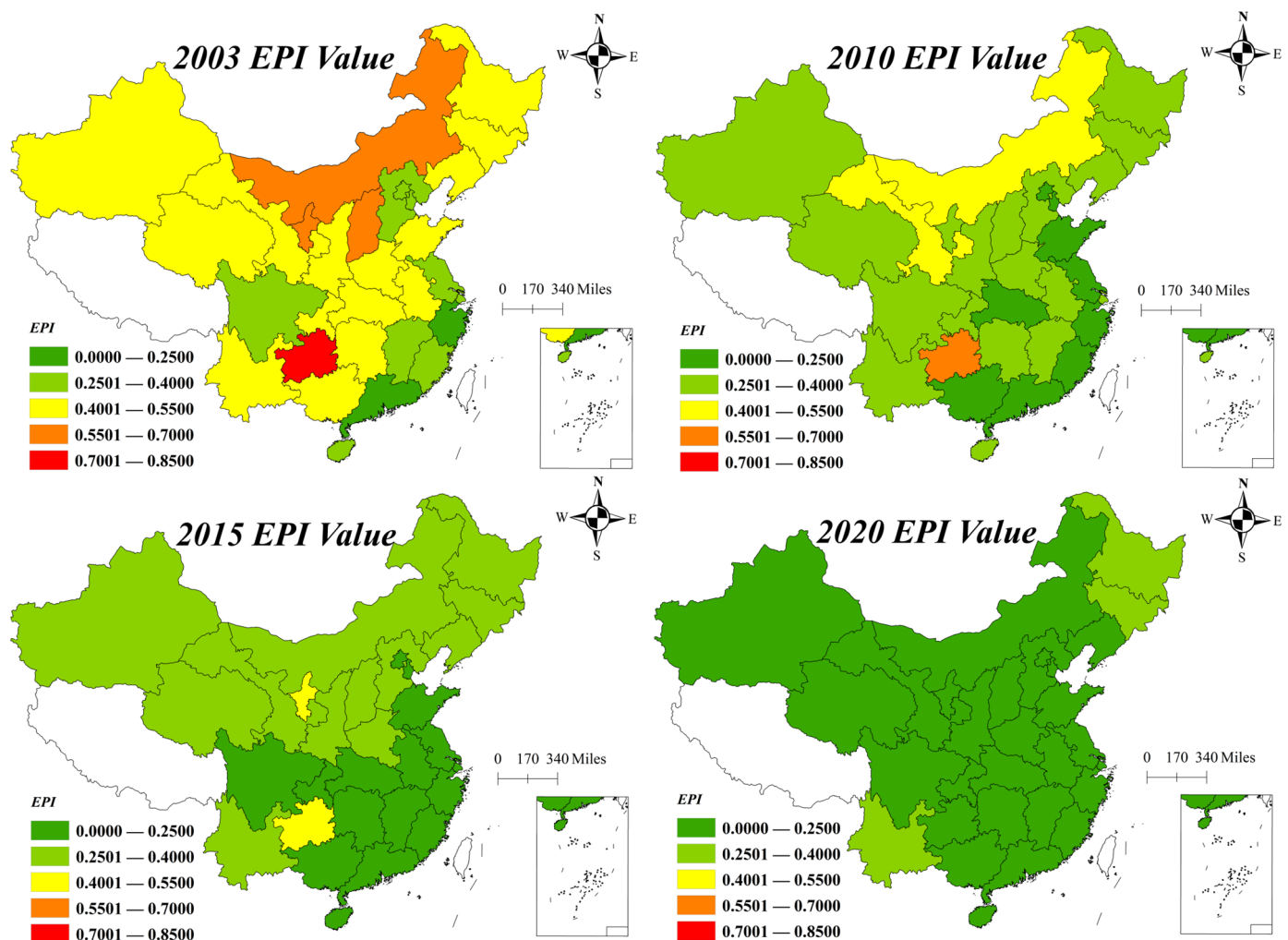


Figure 6. Heat maps of EPI values for specific years.

The heat map in Figure 6 shows clear spatial and temporal differences in EPI values. In 2003, higher EPI levels are concentrated in many western provinces, whereas by 2020 the distribution shifts downward in many provinces, with the largest decreases observed in several southeastern coastal areas. Overall, the maps provide a descriptive visualization of the spatiotemporal evolution of provincial EPI levels over the sample period.

Next, we use the Pearson correlation coefficient to examine the statistical associations between the influencing factors in Table 3 and the EPI, as shown in Figure 7. The correlation heatmap summarizes the direction (positive/negative) and strength of pairwise correlations among variables and the EPI. These results are reported as descriptive evidence of linear associations and are used as background information for the subsequent SHAP-based model interpretation, rather than as evidence of causal effects [52].

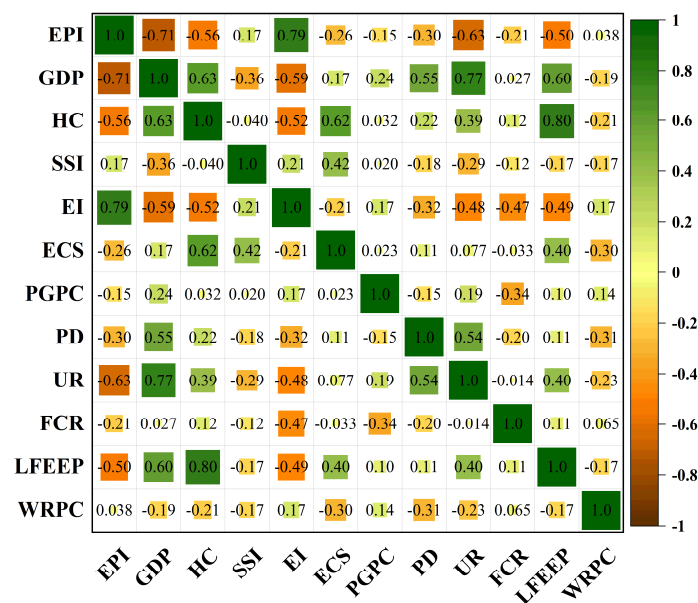


Figure 7. Pearson correlation matrix of influencing factors for EPI.

As shown in Figure 7, several variables are negatively correlated with the EPI. For example, EPI is negatively correlated with GDP (−0.71), HC (−0.56), and ECS (−0.26), indicating that higher values of these variables are associated with lower EPI levels in the sample. By contrast, EI shows a strong positive correlation with EPI (0.79), indicating that higher energy intensity is associated with higher EPI values. UR is negatively correlated with EPI (−0.63), indicating a negative correlation between urbanization and EPI in the observed data. In addition, FCR (−0.21) and LFEED (−0.50) are also negatively correlated with EPI, indicating that higher forest coverage and higher environmental protection expenditure are associated with lower EPI values. Overall, the correlation results provide a descriptive summary of linear associations, which can be compared with the SHAP-based model interpretation and the TWFE regression results reported in later sections.

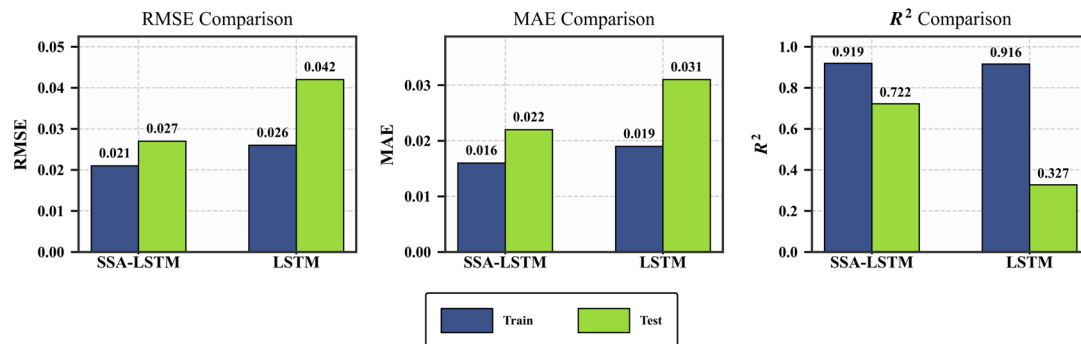
### 3.2. SSA Model Optimization Results

After constructing the Energy Poverty Index (EPI) and its influencing framework, we employed the SSA-LSTM model for energy poverty prediction, with parameter settings provided in Appendix A, Table A1. To ensure a temporally consistent evaluation, the dataset was split into training, validation, and testing subsets using fixed chronological target-year ranges (training: 2008–2017, validation: 2018–2019, and testing: 2020–2022). To avoid the issue of contemporaneous interpretation, we applied a lag to the independent variables, specifically lagging them by three periods. The SSA was optimized using the RMSE of the validation set as the fitness function, and the model was evaluated based on RMSE, MAE, and  $R^2$  metrics [38].

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (15)$$



where *RMSE* represents the average error between the predicted and actual values, with smaller values indicating better performance. *MAE* denotes the average absolute prediction error, where smaller values indicate higher accuracy.  $R^2$  reflects the degree of fit of the model to the data, with larger values being preferable; typically, a value close to 1 indicates a good fit. The evaluation results for the SSA-LSTM model and the LSTM model are shown in Figure 8.



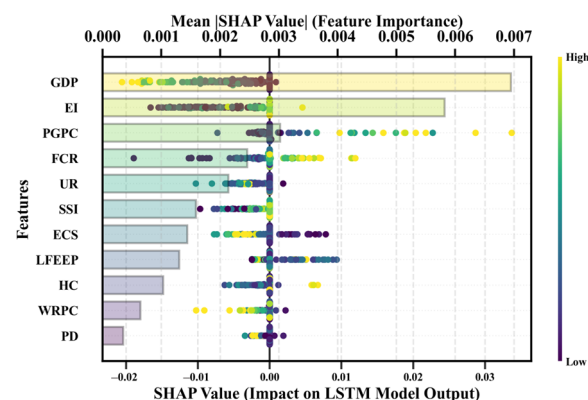
**Figure 8.** Evaluation of SSA-LSTM model.

The results in Figure 8 indicate that the SSA-optimized LSTM model significantly improved prediction accuracy. On the training set,  $R^2$  increased from 0.916 to 0.919, suggesting that the model has learned more valuable features and is better fitting the data. Similarly, on the validation set, the SSA-optimized LSTM demonstrated a notable performance improvement, with RMSE decreasing from 0.042 to 0.027 and MAE decreasing from 0.031 to 0.022, indicating that the optimization method effectively enhanced prediction accuracy.

### 3.3. Results of Identifying Influencing Factors Using the SHAP-LSTM Model

#### 3.3.1. Benchmark Results of SHAP-LSTM

We used the SHAP-LSTM framework to interpret the prediction results of the LSTM model, whose parameters were optimized by the SSA (Figure 9). Through SHAP-based interpretability analysis, the contribution of each feature to the model's predictions can be quantified and compared. Figure 9 presents the relative importance of features and their contributions to the model output. Features such as GDP, EI, and PGPC exhibit relatively large SHAP values, indicating that they make comparatively stronger contributions to the model's prediction of the energy poverty index. By contrast, PD and FCR show smaller SHAP values, suggesting that their contributions to the model prediction are relatively limited [16].



**Figure 9.** SHAP-LSTM analysis results.

In this way, the SHAP-LSTM framework not only improves the interpretability of the prediction results, but also highlights the relative importance of different predictors in the model output, thereby providing useful evidence for understanding model behavior and informing subsequent empirical analysis.

### 3.3.2. Two-Way Fixed Effects Results

Table 4 reports the two-way fixed effects regression results with province and year fixed effects under three lag specifications (independent variables lag phase 1, lag phase 2 and lag phase 3). The association patterns are highly consistent across lag choices. Specifically, EI is positively associated with EPI and remains statistically significant in all specifications, whereas SSI, ECS, PGPC, and UR are negatively associated with EPI and are statistically significant across the three lag models. GDP is also negative and significant across lags, while PD and FCR are positive and significant in all specifications. WRPC shows a negative and significant association with EPI. In contrast, HC is not statistically significant, and LFEEP is only weakly significant in the lag-3 specification. Overall, these fixed-effects estimates indicate stable within-panel associations between EPI and multiple economic, energy, demographic, and environmental covariates when controlling for province and year fixed effects.

**Table 4.** Two-way fixed effects regression results for provincial energy poverty index.

| Variables | Lag Phase 1                |                       | Lag Phase 2                |                       | Lag Phase 3                |                       |
|-----------|----------------------------|-----------------------|----------------------------|-----------------------|----------------------------|-----------------------|
|           | Coef                       | StdErr                | Coef                       | StdErr                | Coef                       | StdErr                |
| Constant  | $3.28 \times 10^{-1} ***$  | $2.42 \times 10^{-2}$ | $3.26 \times 10^{-1} ***$  | $2.45 \times 10^{-2}$ | $3.09 \times 10^{-1} ***$  | $2.48 \times 10^{-2}$ |
| GDP       | $-8.16 \times 10^{-7} ***$ | $1.64 \times 10^{-7}$ | $-8.35 \times 10^{-7} ***$ | $1.86 \times 10^{-7}$ | $-8.83 \times 10^{-7} ***$ | $2.11 \times 10^{-7}$ |
| HC        | $-2.74 \times 10^{-8}$     | $6.28 \times 10^{-7}$ | $-2.75 \times 10^{-7}$     | $7.49 \times 10^{-7}$ | $-7.56 \times 10^{-7}$     | $8.86 \times 10^{-7}$ |
| SSI       | $-7.91 \times 10^{-2} **$  | $3.09 \times 10^{-2}$ | $-8.90 \times 10^{-2} ***$ | $3.09 \times 10^{-2}$ | $-8.66 \times 10^{-2} ***$ | $3.13 \times 10^{-2}$ |
| EI        | $1.05 \times 10^{-1} ***$  | $7.42 \times 10^{-3}$ | $9.77 \times 10^{-2} ***$  | $7.78 \times 10^{-3}$ | $9.09 \times 10^{-2} ***$  | $8.09 \times 10^{-3}$ |
| ECS       | $-5.46 \times 10^{-1} ***$ | $1.73 \times 10^{-1}$ | $-4.78 \times 10^{-1} ***$ | $1.64 \times 10^{-1}$ | $-4.02 \times 10^{-1} **$  | $1.56 \times 10^{-1}$ |
| PGPC      | $-2.53 \times 10^{-2} ***$ | $9.66 \times 10^{-3}$ | $-2.64 \times 10^{-2} ***$ | $1.01 \times 10^{-2}$ | $-2.42 \times 10^{-2} **$  | $1.06 \times 10^{-2}$ |
| PD        | $2.01 \times 10^{-5} ***$  | $4.82 \times 10^{-6}$ | $1.99 \times 10^{-5} ***$  | $4.85 \times 10^{-6}$ | $2.00 \times 10^{-5} ***$  | $4.95 \times 10^{-6}$ |
| UR        | $-1.82 \times 10^{-3} ***$ | $3.20 \times 10^{-4}$ | $-1.81 \times 10^{-3} ***$ | $3.14 \times 10^{-4}$ | $-1.67 \times 10^{-3} ***$ | $3.08 \times 10^{-4}$ |
| FCR       | $6.69 \times 10^{-4} ***$  | $1.65 \times 10^{-4}$ | $6.49 \times 10^{-4} ***$  | $1.65 \times 10^{-4}$ | $6.93 \times 10^{-4} ***$  | $1.71 \times 10^{-4}$ |
| LFEEP     | $5.92 \times 10^{-5}$      | $4.83 \times 10^{-5}$ | $7.11 \times 10^{-5}$      | $5.57 \times 10^{-5}$ | $1.06 \times 10^{-4} *$    | $6.44 \times 10^{-5}$ |
| WRPC      | $-7.65 \times 10^{-6} ***$ | $1.24 \times 10^{-6}$ | $-7.51 \times 10^{-6} ***$ | $1.33 \times 10^{-6}$ | $-7.26 \times 10^{-6} ***$ | $1.41 \times 10^{-6}$ |

Statistical significance is indicated as follows: \*  $p$ -value < 0.1, \*\*  $p$ -value < 0.05, \*\*\*  $p$ -value < 0.01.

Compared with the SHAP-based model interpretation, the TWFE results show broadly consistent directional patterns for several key variables highlighted by SHAP as important contributors to model-predicted EPI variation, including GDP, EI, PGPC, and UR. At the same time, some variables that exhibit relatively smaller SHAP contributions (e.g., PD and FCR) are statistically significant in the TWFE models, suggesting that feature contribution in the nonlinear prediction model and within-panel association in the fixed-effects regression can differ in magnitude. Taken together, the SHAP interpretation and TWFE regression provide complementary perspectives on factor relevance, with SHAP focusing on contributions to model predictions and TWFE summarizing within-panel associations under fixed effects.

### 3.3.3. Heterogeneity Analysis of SHAP-LSTM

To examine regional heterogeneity in model interpretation, we conducted a SHAP-based comparison for China's central, eastern, and western regions. The results are shown in Figure 10. The SHAP results indicate clear regional differences in feature contributions to the LSTM model's predictions. Specifically, in the central region, GDP, EI, and PGPC exhibit relatively large SHAP values, indicating stronger contributions to the model output in this region. In the eastern region, PGPC and GDP remain prominent contributors, while

LFEEP also shows comparatively large SHAP values, suggesting that these predictors are more important in the model interpretation for the eastern provinces. In the western region, UR and PGPC show relatively strong SHAP contributions, with the urbanization-related predictor displaying high importance in the regional prediction results. By contrast, EI and ECS show comparatively smaller SHAP values in the western region, indicating weaker contributions to the model prediction in this subsample [25].

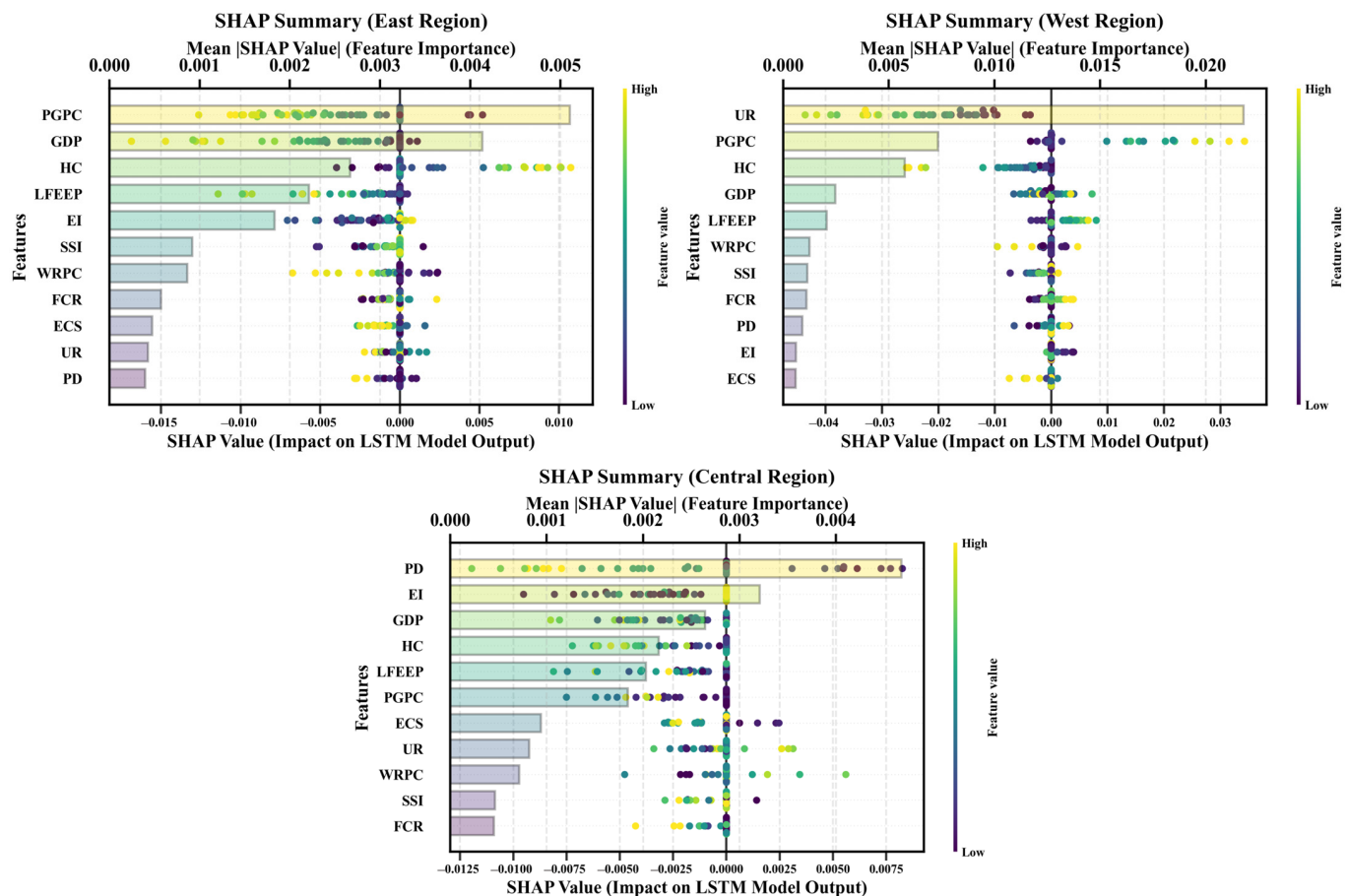


Figure 10. Results of SHAP-LSTM heterogeneity analysis.

These regional heterogeneity results provide additional evidence that the relative importance of predictors in the SHAP-LSTM interpretation varies across regions. This finding highlights the value of conducting region-specific model interpretation when analyzing provincial energy poverty prediction patterns.

### 3.3.4. Sensitivity Analysis for Potential Index-Construction Overlap

To assess the robustness of the SHAP-LSTM interpretation under alternative index construction choices, this study conducts a sensitivity analysis based on an alternative specification of the dependent variable. Specifically, the “no urban” specification denotes a reconstructed energy poverty index (EPI) in which urban-specific indicators are removed from the entropy-weighted indicator system, including the urban natural gas penetration rate, urban per capita natural gas supply, refrigerators per 100 urban households, and air conditioners per 100 urban households. Using this alternative EPI, the SHAP-LSTM analysis is re-estimated under the same modeling framework and parameter settings. This procedure enables a direct comparison of feature-importance patterns across index specifications and helps evaluate whether the contribution of the urbanization rate (UR) is sensitive to the inclusion of urban-related EPI components.

Figure 11 compares SHAP results from the baseline composite EPI (left) and the no urban EPI specification (right). Overall, the importance structure remains broadly stable: key variables such as GDP and EI continue to rank among the most influential predictors, indicating that the model's main predictive patterns are not fundamentally altered by excluding urban-specific EPI components. At the same time, the ranking and relative importance of several mid-level variables shift, and the importance of UR is moderately attenuated in the no urban specification while remaining non-zero and still within the middle importance tier. This pattern suggests that part of the UR contribution in the baseline model may reflect index-construction overlap, but the UR–EPI association is not entirely a tautological artifact. Therefore, the sensitivity results support a more cautious interpretation of UR as a robust predictive correlate of model-predicted EPI, rather than a direct causal mechanism.

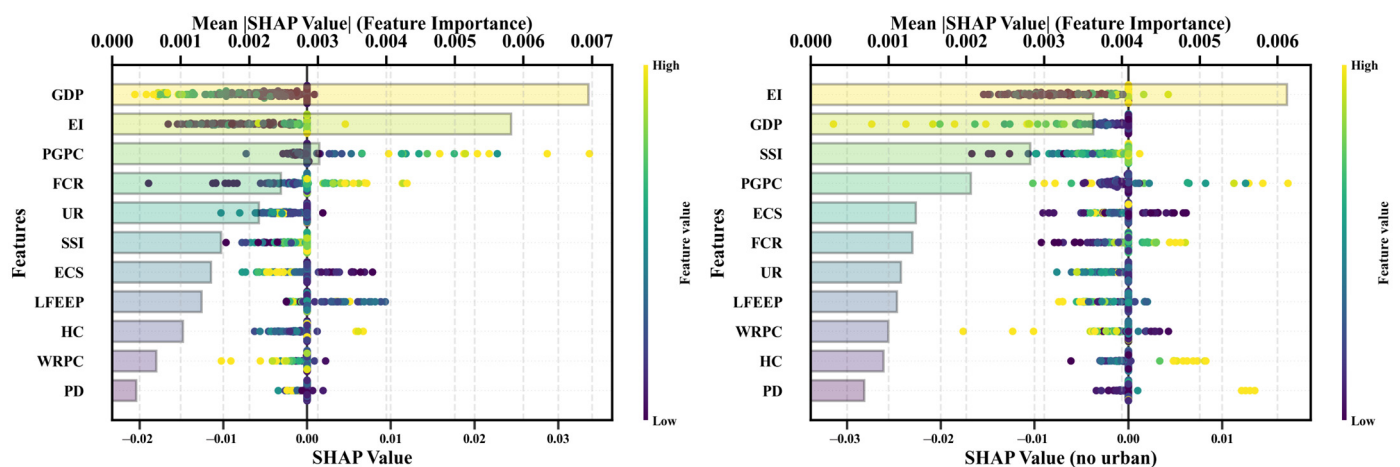


Figure 11. Sensitivity Analysis for Potential Index-Construction Overlap.

### 3.4. The Prediction Results of SSA-LSTM

We used the SSA-LSTM model for prediction, and the results are shown in Figure 12. The figure illustrates the model's fit to the actual values, with an  $R^2$  value of 0.722, indicating strong predictive capability. In contrast, the LSTM model's  $R^2$  value is 0.327, suggesting a poorer fit and further validating the effectiveness of the SSA optimization [37].

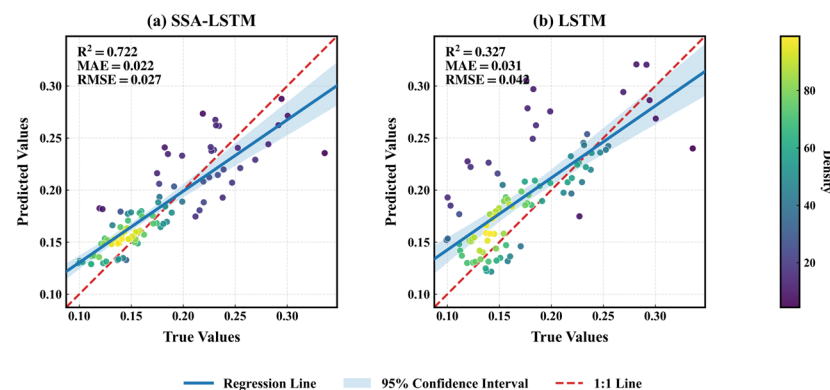


Figure 12. SSA-LSTM prediction results.

## 4. Discussion

In Section 4, we will conduct four main analyses. First, we use the SHAP-LSTM model to perform a Top Feature Dependencies Analysis on the top six features; second, we analyze the robustness of the SHAP-LSTM model; third, we compare the SSA-LSTM model with

other models to explore its performance; and fourth, we analyze the convergence of the SSA model in this study and compare it with other models.

#### 4.1. Top Feature Dependencies Identified by SHAP-LSTM

Top Feature Dependencies Analysis is used to illustrate how the feature values are associated with their SHAP contributions to the model output, thereby providing a detailed interpretation of the prediction behavior. Using SHAP values, we quantify each feature's contribution to the model-predicted EPI at the sample level. Specifically, the dependencies are visualized by plotting the SHAP values of a feature against the corresponding feature values, which helps characterize how the contribution pattern changes across different value ranges (Figure 13). To summarize the overall tendency, fitted trend lines are used to describe the relationship between feature values and SHAP contributions. The red dashed line denotes an identified threshold reference point, and the black dashed line indicates the median value, which facilitates the visual inspection of potential turning points in the SHAP contribution patterns [22].

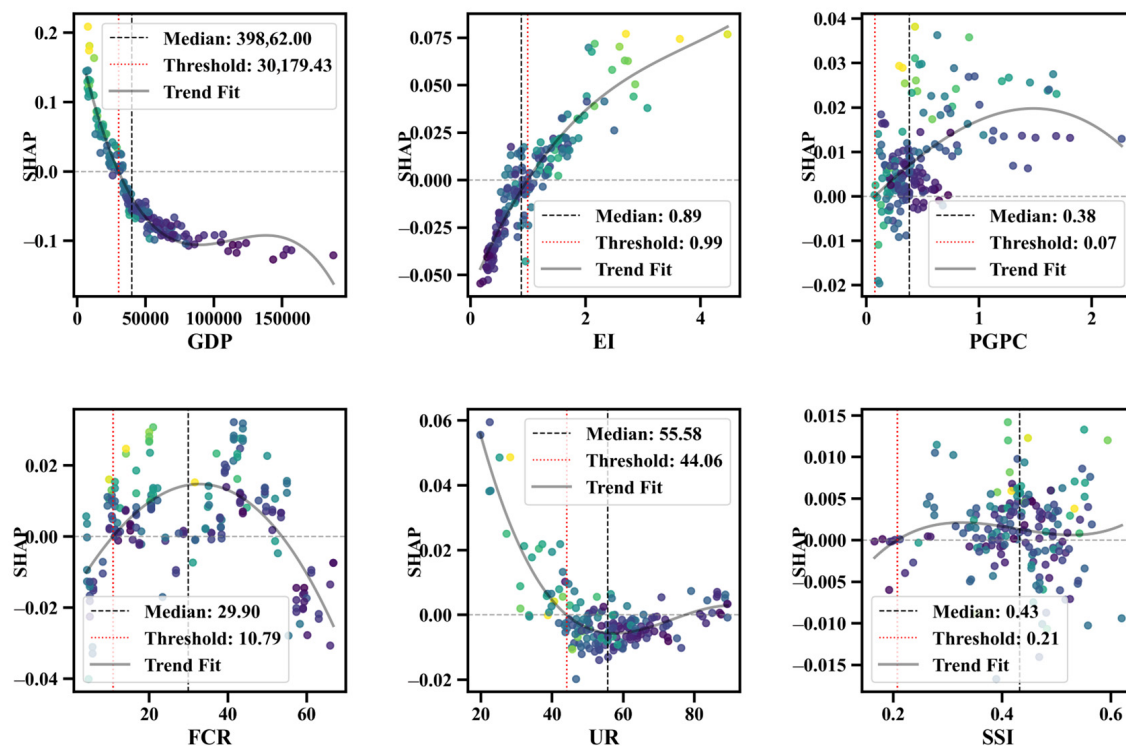


Figure 13. Top dependencies of the first six features in SHAP-LSTM.

Figure 13 shows the dependency patterns for the top six features. Overall, the relationship between feature values and SHAP contributions differs across variables. For GDP, the SHAP dependence pattern becomes increasingly negative as GDP rises, indicating that higher GDP levels are generally associated with lower model-predicted EPI (i.e., negative SHAP contributions in the higher-GDP range). In contrast, EI exhibits predominantly positive SHAP contributions that increase with EI, suggesting that higher energy intensity is associated with higher model-predicted EPI. These directions are broadly consistent with the Pearson correlation patterns reported in Figure 7. In addition, PGPC, FCR, and SSI display nonlinear SHAP dependence patterns, indicating that their contributions to the model prediction vary across value ranges and may change more noticeably around certain reference points. UR also shows a nonlinear dependence pattern, implying that its contribution to model-predicted EPI is not constant across the distribution. Overall, the



dependency analysis provides a more fine-grained interpretation of how key predictors contribute to the model output across different feature levels, complementing the global feature-importance results.

#### 4.2. Robustness Analysis of SHAP-LSTM Model

To further examine whether the indicator-system design and weighting method affect the downstream interpretation results, we reconstructed the EPI using multiple alternative weighting methods and conducted a robustness comparison. We employed various EPI calculation models, including CRITIC, the Coefficient of Variation (CV), and the Maximum Difference Method (DM), and ran SHAP-LSTM for analysis on each. By comparing the ranking of influencing factors across different models, we evaluate whether the SHAP-LSTM model maintains a consistent feature importance ranking. As shown in Figure 14, despite using different EPI calculation methods, the top two most important influencing factors in the SHAP-LSTM model remain GDP and EI, with their SHAP values significantly higher than those of other factors. This result indicates that the SHAP-LSTM model maintains high robustness across different EPI calculation models, confirming the reliability and consistency of the model [34].

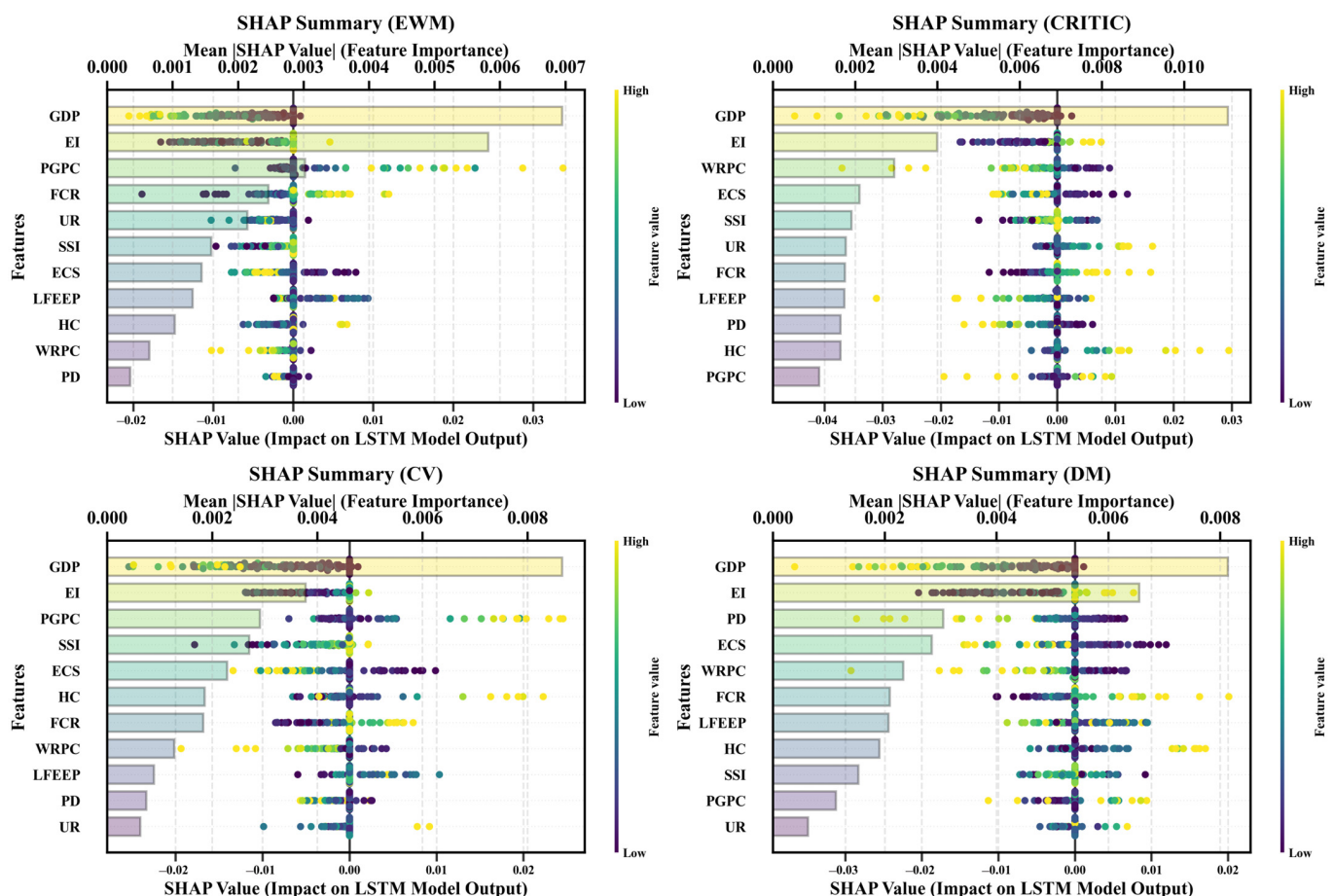


Figure 14. Robustness analysis of SHAP-LSTM model.

#### 4.3. Comparison of SSA-LSTM Model with Other Popular Models

In this section, we compare the SSA-LSTM model with several popular deep learning and machine learning models. The selected comparison models include LSTM [49], Transformer [53], and ANN [54], among others. All models were trained and tested with the same parameter configurations to ensure a fair comparison, with parameter settings

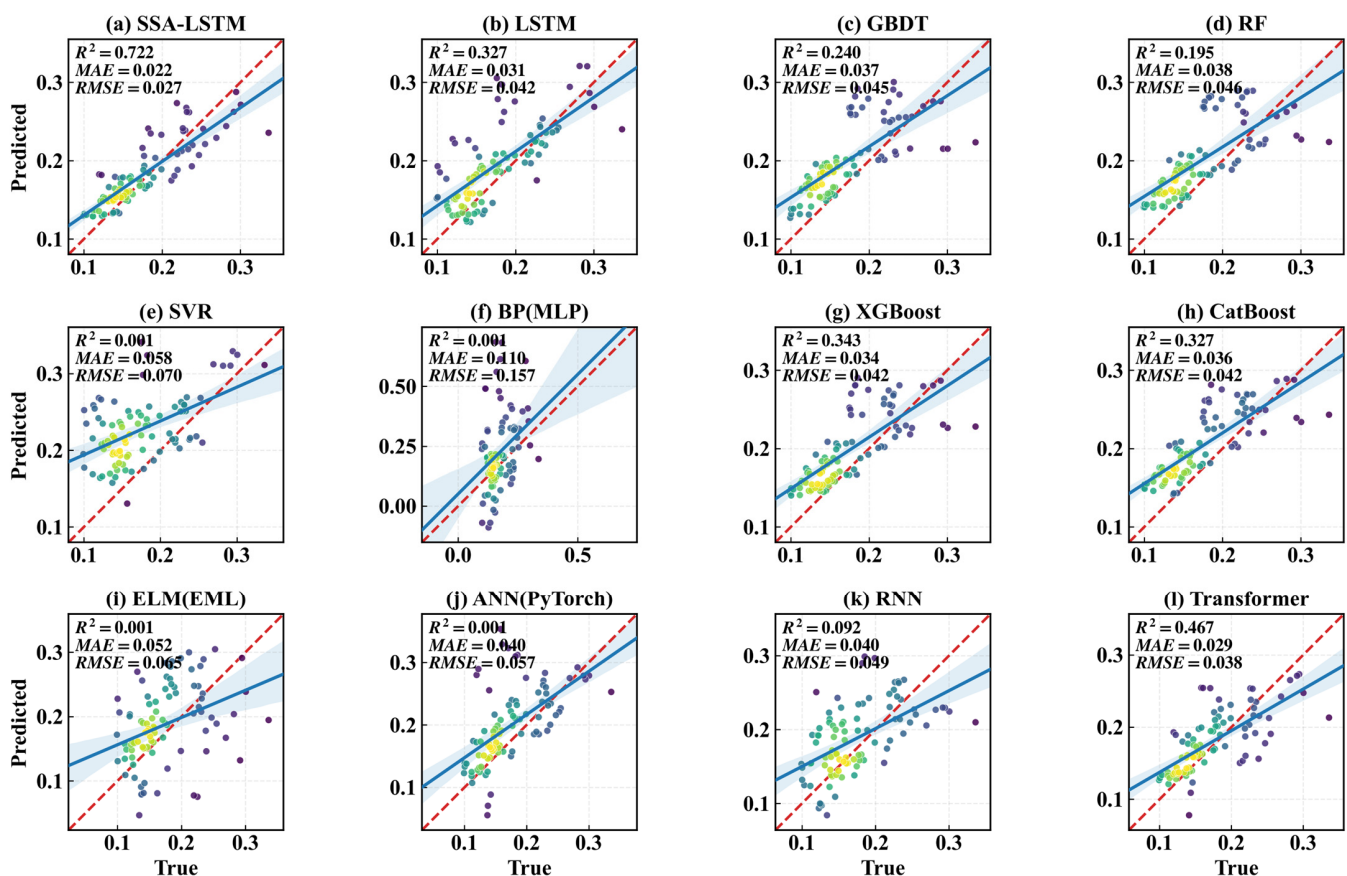


provided in Appendix A, Table A1. The evaluation results are shown in Table 5, and the prediction results are displayed in Figure 15.

**Table 5.** Comparison of SSA-LSTM and benchmark models for energy poverty prediction.

| Model       | Training Dataset |       |                | Testing Dataset |       |                |
|-------------|------------------|-------|----------------|-----------------|-------|----------------|
|             | RMSE             | MAE   | R <sup>2</sup> | RMSE            | MAE   | R <sup>2</sup> |
| SSA-LSTM    | 0.021            | 0.016 | 0.919          | 0.027           | 0.022 | 0.722          |
| LSTM        | 0.026            | 0.019 | 0.916          | 0.042           | 0.031 | 0.327          |
| GBDT        | 0.009            | 0.007 | 0.990          | 0.045           | 0.037 | 0.240          |
| RF          | 0.012            | 0.009 | 0.983          | 0.046           | 0.038 | 0.195          |
| SVR         | 0.008            | 0.008 | 0.992          | 0.070           | 0.058 | 0.001          |
| BP          | 0.024            | 0.019 | 0.929          | 0.157           | 0.110 | 0.001          |
| XGBoost     | 0.000            | 0.000 | 1.000          | 0.042           | 0.034 | 0.343          |
| CatBoost    | 0.001            | 0.000 | 1.000          | 0.042           | 0.036 | 0.327          |
| ELM         | 0.000            | 0.000 | 1.000          | 0.065           | 0.052 | 0.001          |
| ANN         | 0.011            | 0.008 | 0.986          | 0.057           | 0.040 | 0.001          |
| RNN         | 0.043            | 0.034 | 0.778          | 0.049           | 0.040 | 0.092          |
| Transformer | 0.018            | 0.014 | 0.926          | 0.038           | 0.029 | 0.467          |

In Table 5, R<sup>2</sup> values below zero are displayed as 0.001 for readability.



**Figure 15.** Multimodel Prediction Results.

As shown in Table 5, the SSA-LSTM model achieves the best overall predictive performance on the testing set, with RMSE = 0.027, MAE = 0.022, and R<sup>2</sup> = 0.722. Compared with the baseline LSTM model (RMSE = 0.042, MAE = 0.031, R<sup>2</sup> = 0.327), the SSA-LSTM substantially improves prediction accuracy and generalization performance, indicating that SSA-based hyperparameter optimization effectively enhances the temporal learning capability of the LSTM framework for provincial energy poverty prediction. The SSA-

LSTM model also outperforms other benchmark models, including XGBoost ( $R^2 = 0.343$ ), CatBoost ( $R^2 = 0.327$ ), and Transformer ( $R^2 = 0.467$ ), confirming its stronger robustness under the current prediction setting.

A comparison of training and testing results further shows that several benchmark models (e.g., GBDT, RF, XGBoost, CatBoost, and ELM) obtain very high training performance but exhibit substantial performance deterioration on the testing set, suggesting a higher risk of overfitting under the time-ordered prediction task.

Figure 15 presents the prediction results of multiple models. The SSA-LSTM model clearly outperforms the others in terms of how closely its predictions align with the actual values. This further confirms the model's superiority in accurately forecasting energy poverty, demonstrating its ability to capture complex patterns and trends more effectively than the other models.

#### 4.4. Comparison of SSA with Other Search Algorithms and Population Parameters

In this section, we compare the convergence speed and model performance of SSA with several other search algorithms, including Genetic Algorithm (GA) [55], Whale Optimization Algorithm (WOA) [47], Simulated Annealing (SA) [56], and Particle Swarm Optimization (PSO) [57]. Using the same parameter settings, we analyze the convergence characteristics and improvement effects of each algorithm, with the parameter settings provided in Appendix A, Table A1. Additionally, we performed a robustness analysis of SSA's population parameters by adjusting the population sizes to 25, 50, 75, 100, and 125, evaluating their effects on convergence speed and search capability [47].

Figure 16 results indicate that the SSA has the fastest convergence speed and achieves the lowest RMSE value. It successfully escapes local optima after 120 iterations, further optimizing the model. In contrast, WOA converges more slowly in the initial stages but escapes local optima around the 80th iteration, demonstrating a faster convergence speed than GA. Both SA and PSO get trapped in local optima at an early stage, converging too quickly and failing to further improve model performance.

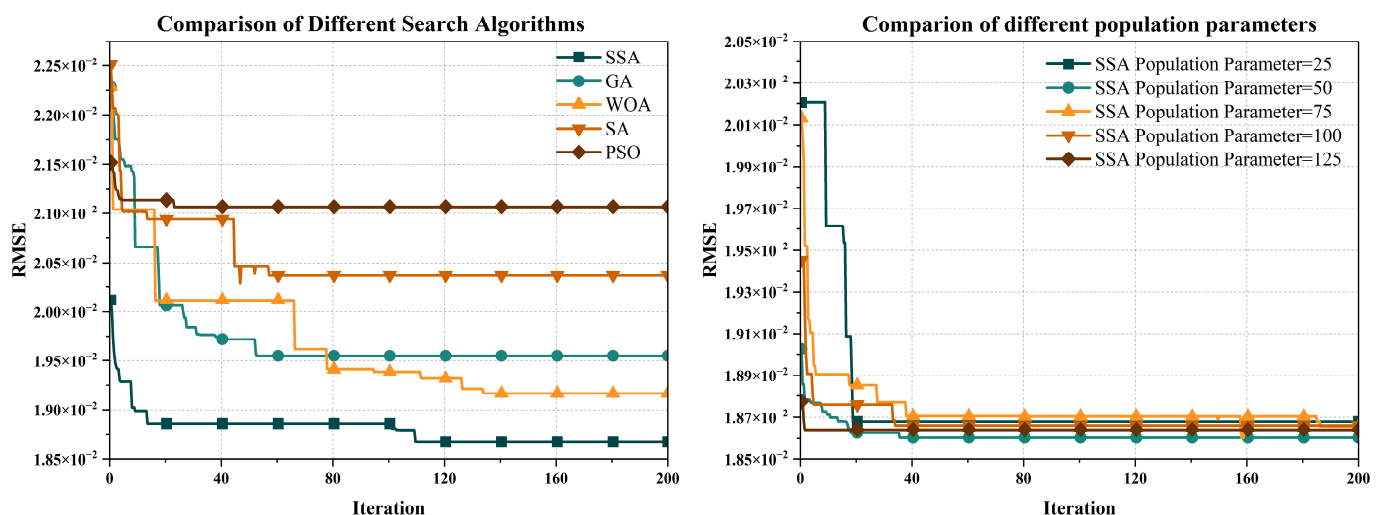


Figure 16. Convergence analysis of SSA.

In the comparison of population parameters, the SSA model performs best in terms of both convergence speed and search capability when the population size is set to 50. Smaller populations, such as 25, result in slower convergence, while larger populations tend to fall into local optima more easily, exhibiting less stable search capability compared to the population size of 50.

## 5. Conclusions

This study develops an SSA-LSTM-based forecasting and interpretation framework to improve provincial energy poverty prediction. The results show that the proposed SSA-LSTM model achieves superior predictive performance relative to benchmark models under the chronologically ordered evaluation setting, with clear advantages in RMSE, MAE, and  $R^2$  on the testing set, indicating improved out-of-sample prediction capability. The SHAP-based model interpretation further shows that variables such as GDP, EI, and PGPC make relatively strong contributions to model-predicted EPI variation, while the robustness analyses indicate that the main interpretation patterns remain stable across alternative EPI construction methods and sensitivity settings. In addition, the two-way fixed effects (TWFE) regression results provide complementary evidence: EI is positively associated with EPI, whereas SSI, ECS, PGPC, and UR are negatively associated with EPI, and several key variables highlighted in the SHAP analysis (e.g., GDP, EI, PGPC, and UR) also appear statistically significant in the panel regression results [58].

These findings provide several practical implications for policymakers. First, the SSA-LSTM framework can be used as a provincial early-warning and monitoring tool to identify provinces with elevated predicted EPI levels and to support the prioritization of intervention timing and resource allocation. Second, the combined SHAP and TWFE results suggest the importance of differentiated policy strategies across regions: for provinces where EI and energy structure indicators are more prominent, policy attention may focus on energy efficiency improvement and structural optimization; for regions where PGPC, UR, or infrastructure-related indicators are more important, policy efforts may prioritize energy supply capacity, grid/infrastructure upgrading, and access-related improvements. Third, policymakers can use the model outputs together with key predictor monitoring (e.g., EI, UR, PGPC, and related economic indicators) to build a more targeted and data-informed policy assessment framework for provincial energy poverty governance.

Overall, this study contributes a provincial panel-based framework that combines prediction performance evaluation, SHAP-based model interpretation, and TWFE-based empirical analysis. This integrated design improves the transparency and practical interpretability of energy poverty prediction compared with using prediction models alone. At the same time, several limitations remain. The analysis is based on historical data and does not explicitly incorporate future policy scenarios or external shocks. Future research can extend this framework by introducing scenario simulation, richer explanatory variables, and additional robustness checks under alternative policy and macroeconomic conditions.

**Author Contributions:** Conceptualization, Z.F. and P.F.; methodology, Z.F.; software, Z.F.; validation, P.F., Z.F. and Y.W.; formal analysis, Y.W.; investigation, Y.W.; resources, Y.W.; data curation, Y.W.; writing—original draft preparation, Z.F.; writing—review and editing, Z.F., P.F. and Y.W.; visualization, P.F.; supervision, Y.W.; project administration, Y.W.; funding acquisition, Y.W. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research was funded by National Social Science Fund of China, grant number 22BJY259.

**Data Availability Statement:** The data supporting the reported results in this study are available upon request from the corresponding author.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## Appendix A

**Table A1.** The hyperparameters of the algorithms used in this paper.

| Model       | Parameters                                                                                                    |
|-------------|---------------------------------------------------------------------------------------------------------------|
| LSTM        | Hidden units: 64, Learning rate: 0.001, Batch size: 32, Epoch: 500                                            |
| Transformer | Layers: 6, Hidden units: 128, Learning rate: 0.001, Epoch: 500                                                |
| ANN         | Layers: 3, Hidden units: 64, Learning rate: 0.01, Batch size: 32, Epoch: 500                                  |
| RNN         | Hidden units: 64, Learning rate: 0.001, Batch size: 32, Epoch: 500                                            |
| GRU         | Hidden units: 64, Learning rate: 0.001, Batch size: 32, Epoch: 500                                            |
| CatBoost    | Learning rate: 0.1, Iterations: 500, Depth: 6                                                                 |
| XGBoost     | Learning rate: 0.1, Max depth: 6, Iterations: 500                                                             |
| GBDT        | Learning rate: 0.1, Max depth: 6, Iterations: 500                                                             |
| RF          | Trees: 100, Max depth: 6, Min samples split: 2                                                                |
| SVR         | C: 1, Kernel: 'rbf', Gamma: 'scale'                                                                           |
| ELM         | Hidden units: 1000, Activation function: 'sigmoid'                                                            |
| SSA         | Population size: 50, Max iterations: 200, Learning rate: 0.001                                                |
| GA          | Population size: 50, Generations: 200, Crossover rate: 0.8, Mutation rate: 0.1                                |
| WOA         | Population size: 50, Max iterations: 200, Exploration coefficient: 0.5                                        |
| SA          | Initial temperature: 1000, Cooling rate: 0.95, Iterations: 200                                                |
| PSO         | Swarm size: 50, Max iterations: 200, Inertia weight: 0.7, Cognitive coefficient: 1.5, Social coefficient: 1.5 |

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